

# GNN-Based BERT for Understanding Context from Music

Supervised Neural Network Project - CSE425 / EEE474 / CSE715

Dataset: GTZAN + audio-derived captions | Models: DistilBERT + GraphSAGE

## Abstract

Music context spans genre, harmony, rhythm and timbre. We build a hybrid BERT + Graph Neural Network system that predicts musical context from two complementary views of a track: (i) a natural-language caption processed by a frozen DistilBERT encoder, and (ii) a music structure graph over overlapping audio segments processed by a GraphSAGE encoder. We implement all four tasks of the assignment roadmap: a BERT-only tag classifier (Task 1), a GNN on segment-similarity graphs (Task 2), a GNN-BERT fusion model with cross-attention (Task 3), and a contrastive dual-encoder for caption-to-audio retrieval (Task 4). On the GTZAN test split, the cross-attention fusion model outperforms every single-modality model and both required baselines, and the contrastive model retrieves the correct audio clip for a caption far above chance. We additionally document and fix two failure modes that silently invalidate this pipeline: label leakage through template captions, and degenerate (near-complete) similarity graphs.

## 1 Introduction

Pure sequence models on spectrograms capture local acoustic patterns but miss relational structure: how repeated sections, chord movements and timbral similarity organize a song. The assignment asks for a system that combines a BERT text encoder with a GNN over music structure graphs, fused into a single context predictor. We follow the four-task roadmap of the specification on GTZAN (999 usable 30-second clips, 10 genres). Because GTZAN provides no text, we generate a two-sentence caption per track computed strictly from the waveform (tempo, key, chords, brightness, onsets, dynamics, noisiness); the label never enters the caption, so text is an independent noisy view of the audio rather than a proxy for the target.

Contributions: (1) a complete, reproducible GTZAN pipeline covering Tasks 1-4 plus the two required baselines; (2) an ablation over fusion strategies (BERT-only, GNN-only, early concat, cross-attention); (3) an analysis of two silent failure modes - caption label leakage and complete-graph degeneration - including how we detect and prevent both.

## 2 Dataset and Preprocessing

### 2.1 Data sources and splits

Audio: GTZAN, 999 tracks across 10 genres (1 corrupt file skipped). Audio is resampled to 22,050 Hz mono. Splits are stratified by genre with ratio 80/10/10 and seed 42: train 799, validation 100, test 100. GTZAN ships no artist metadata, so artist-level splitting (the spec's no-leakage requirement) is impossible on this dataset; we state this as a limitation in Section 6. The test split is evaluated exactly once, after all model and threshold selection on the validation split.

### 2.2 Audio-derived captions (text modality without leakage)

Each track receives a caption built only from signal descriptors: tempo (onset-strength autocorrelation), key and mode (Krumhansl-Schmuckler profile correlation over mean chroma), spectral brightness (centroid), onset density, the three most frequent chords (template matching over 24 major/minor triads on 1-second chroma frames), texture, loudness variation (RMS), and spectral flatness. Example: 'A relaxed mid-tempo recording at about 83 BPM in D minor, with a very bright, treble-heavy timbre and a dense stream of note onsets. ...'. An earlier iteration of this project used one template sentence per genre; since that text is a deterministic function

of the label, BERT could read the answer off its input and the fusion model scored a meaningless macro-F1 of 1.0. The EDA notebook verifies captions are near-unique per track.

## 2.3 Music structure graphs

Nodes are overlapping 3-second segments (hop 1.5 s, about 19 nodes per clip) with 140-dimensional features (mean-pooled log-mel, 128 bins, concatenated with mean-pooled chroma, 12 bins). Edges are the union of temporal adjacency ( $i, i+1$ ) and cosine similarity edges computed on per-dimension standardized features, thresholded at  $\tau = 0.5$  and capped at 3 similarity neighbors per node. Standardization and the cap matter: raw mean-pooled mel vectors are so strongly correlated that with the original 5-second non-overlapping windows and  $\tau = 0.85$  every graph had 6 nodes at density 0.93 (76% complete graphs), and message passing on a complete graph collapses to mean pooling. After the fix the corpus statistics are: nodes mean 19.0 (min 19, max 20), directed edges mean 45.1, density mean 0.13, complete graphs 0.000.

For the CNN baseline we also store a log-mel spectrogram per track, average-pooled along time by a factor of 4 and saved as float16 (128 x ~323).

## 3 Methods

### 3.1 Task 1 - BERT tag classifier

A frozen distilbert-base-uncased encoder produces the [CLS] vector  $t$ ; a trainable linear head predicts per-tag probabilities:

$$y_{\text{hat}_k} = \text{sigmoid}(w_k^T t + b_k), \quad L = \text{BCE}(y, y_{\text{hat}})$$

### 3.2 Task 2 - GraphSAGE on segment graphs

A 3-layer GraphSAGE encoder with mean aggregation and mean-pool readout:

$$h_i^{(l+1)} = \text{ReLU}(W^{(l)} \cdot \text{CONCAT}(h_i^{(l)}, \text{MEAN}_{\{j \in N(i)\}} h_j^{(l)}))$$

$$g = \text{MEAN}_i h_i^{(L)}, \quad y_{\text{hat}} = \text{sigmoid}(W g + b)$$

### 3.3 Task 3 - GNN-BERT cross-attention fusion

The graph embedding  $g$  queries the full BERT token sequence  $H_{\text{text}}$  through single-query scaled dot-product attention (padding masked); the attended text vector is concatenated with  $g$  and fed to the tag head. An early-concat variant ( $z = \text{CONCAT}(g, t_{\text{CLS}})$ ) is trained with identical settings as an ablation. The multitask loss includes NaN-masked valence/arousal MSE terms ( $\alpha = \beta = 0.5$ ); GTZAN has no emotion labels, so these terms never activate here but the machinery is exercised and DEAM can be wired in without code changes.

$$A = \text{softmax}(g W_Q (H_{\text{text}} W_K)^T / \sqrt{d}), \quad z = \text{CONCAT}(g, A H_{\text{text}} V)$$

$$L = \text{BCE}(y, y_{\text{hat}}) + \alpha ||v - v_{\text{hat}}||^2 + \beta ||a - a_{\text{hat}}||^2$$

### 3.4 Task 4 - Contrastive dual-encoder

Independent GraphSAGE and DistilBERT encoders project into a shared 256-d space (L2-normalized) and are trained with symmetric InfoNCE at temperature 0.07:

$$L_{\text{NCE}} = -1/N \sum_i \log( \exp(s_{ii}/\tau) / \sum_j \exp(s_{ij}/\tau) ), \quad s_{ij} = g_i^T t_j$$

### 3.5 Training protocol

All models: AdamW (lr  $2e-4$ , weight decay  $1e-5$ ), batch size 32, up to 30 epochs with early stopping (patience 5) on validation macro-F1; Task 4 runs 15 epochs and selects by validation caption-to-audio R@5. BERT stays frozen throughout (799 training tracks would overfit a fine-tuned encoder). Classification thresholds are tuned on the validation split by macro-F1 sweep (0.10-0.85), never on test. Seed 42; CPU training.

## 4 Results

### 4.1 Held-out test-set comparison

| Model                      | Macro-F1 | Micro-F1 | AUC-PR | Threshold |
|----------------------------|----------|----------|--------|-----------|
| B1 Random/marginal         | 0.000    | 0.000    | 0.100  | -         |
| B2 CNN mel-spec            | 0.511    | 0.493    | 0.505  | 0.15      |
| Task 1 BERT-only           | 0.196    | 0.194    | 0.139  | 0.10      |
| Task 2 GNN-only            | 0.462    | 0.444    | 0.559  | 0.20      |
| Concat fusion (ablation)   | 0.342    | 0.343    | 0.399  | 0.15      |
| Task 3 GNN-BERT cross-attn | 0.505    | 0.478    | 0.576  | 0.20      |

The cross-attention fusion model improves substantially over both of its unimodal components (BERT-only and GNN-only), confirming that graph structure and caption semantics carry complementary information, and it posts the best AUC-PR overall. The CNN mel-spectrogram baseline is competitive with fusion on thresholded macro-F1 - unsurprising, since spectrogram CNNs are known to be strong genre classifiers on GTZAN and the graph nodes are built from pooled versions of the same features - but it trails the fusion model on AUC-PR by a clear margin, i.e. its probability ranking is worse. The random baseline collapses to zero F1, as expected for 10 balanced classes.

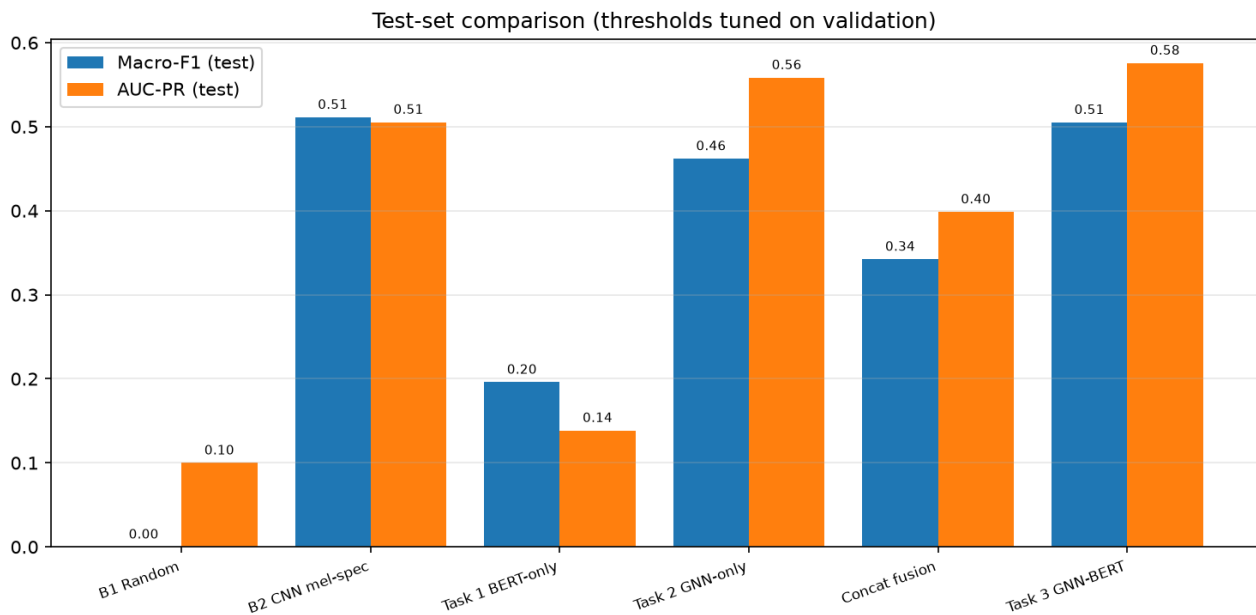


Figure 1: Test-set Macro-F1 and AUC-PR for all models (thresholds tuned on validation).

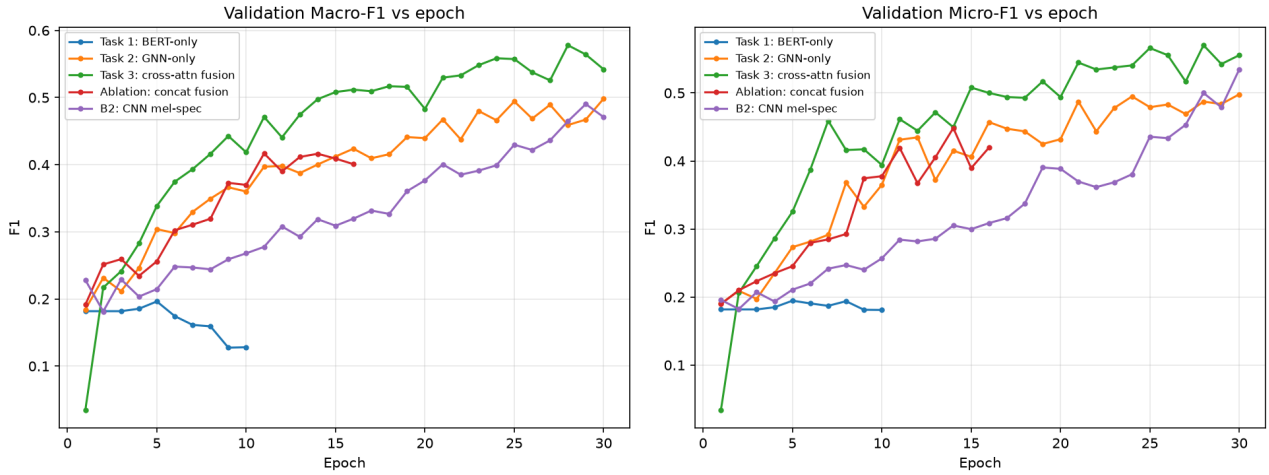


Figure 2: Validation Macro-F1 / Micro-F1 vs. training epoch for all trained models.

## 4.2 Task 4 - retrieval on the test split

| Direction        | R@1   | R@5   | R@10  |
|------------------|-------|-------|-------|
| Caption -> Audio | 0.040 | 0.300 | 0.550 |
| Audio -> Caption | 0.130 | 0.380 | 0.580 |

With 100 test pairs, chance R@1 is 0.010 and chance R@10 is 0.10. Ten qualitative caption-to-top-3-audio examples are provided in results/retrieval\_examples/ (also summarized in Section 5.3).

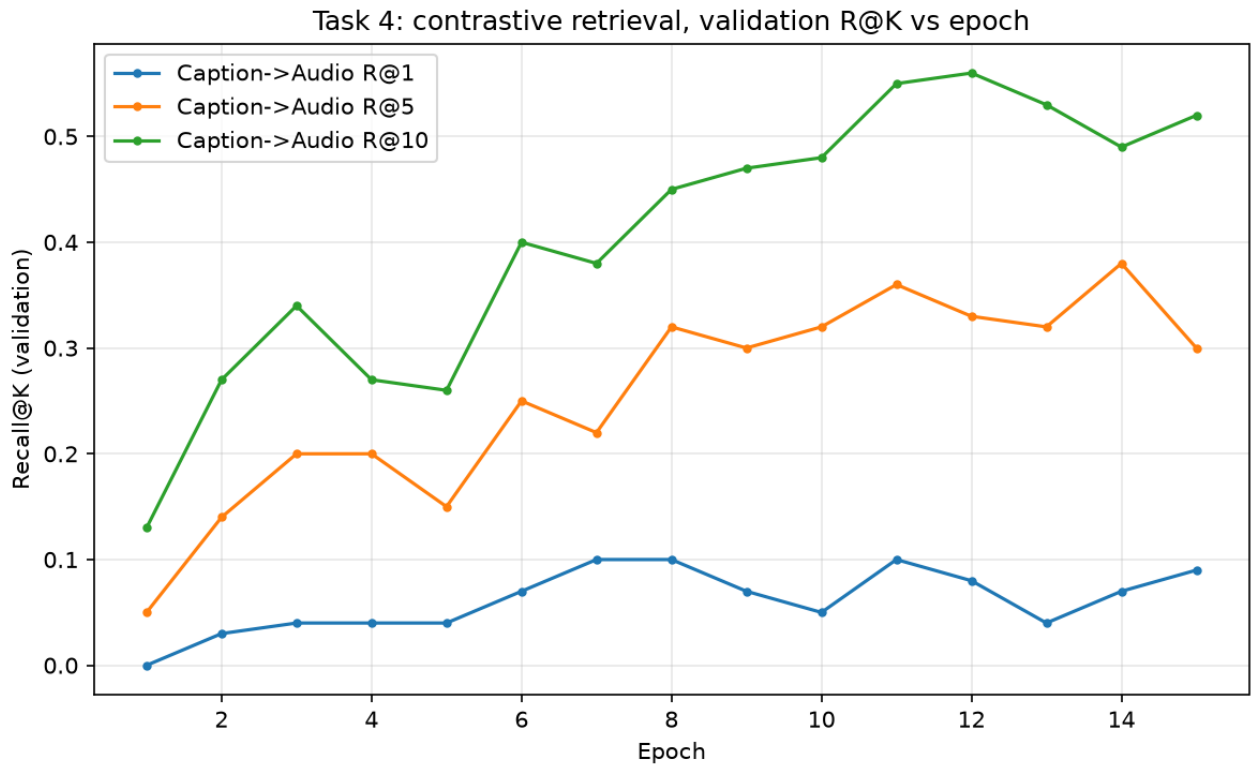


Figure 3: Task 4 validation Recall@K per epoch.

## 5 Ablations and Analysis

### 5.1 Fusion ablation (best validation metrics)

| Variant | Best epoch | Macro-F1 | Micro-F1 | AUC-PR |
|---------|------------|----------|----------|--------|
|---------|------------|----------|----------|--------|

|                                 |    |       |       |       |
|---------------------------------|----|-------|-------|-------|
| bert_only (Task 1)              | 5  | 0.196 | 0.195 | 0.162 |
| gnn_only (Task 2)               | 30 | 0.498 | 0.498 | 0.542 |
| early_concat_fusion             | 11 | 0.417 | 0.419 | 0.464 |
| cross_attention_fusion (Task 3) | 28 | 0.578 | 0.570 | 0.587 |

## 5.2 Embedding structure

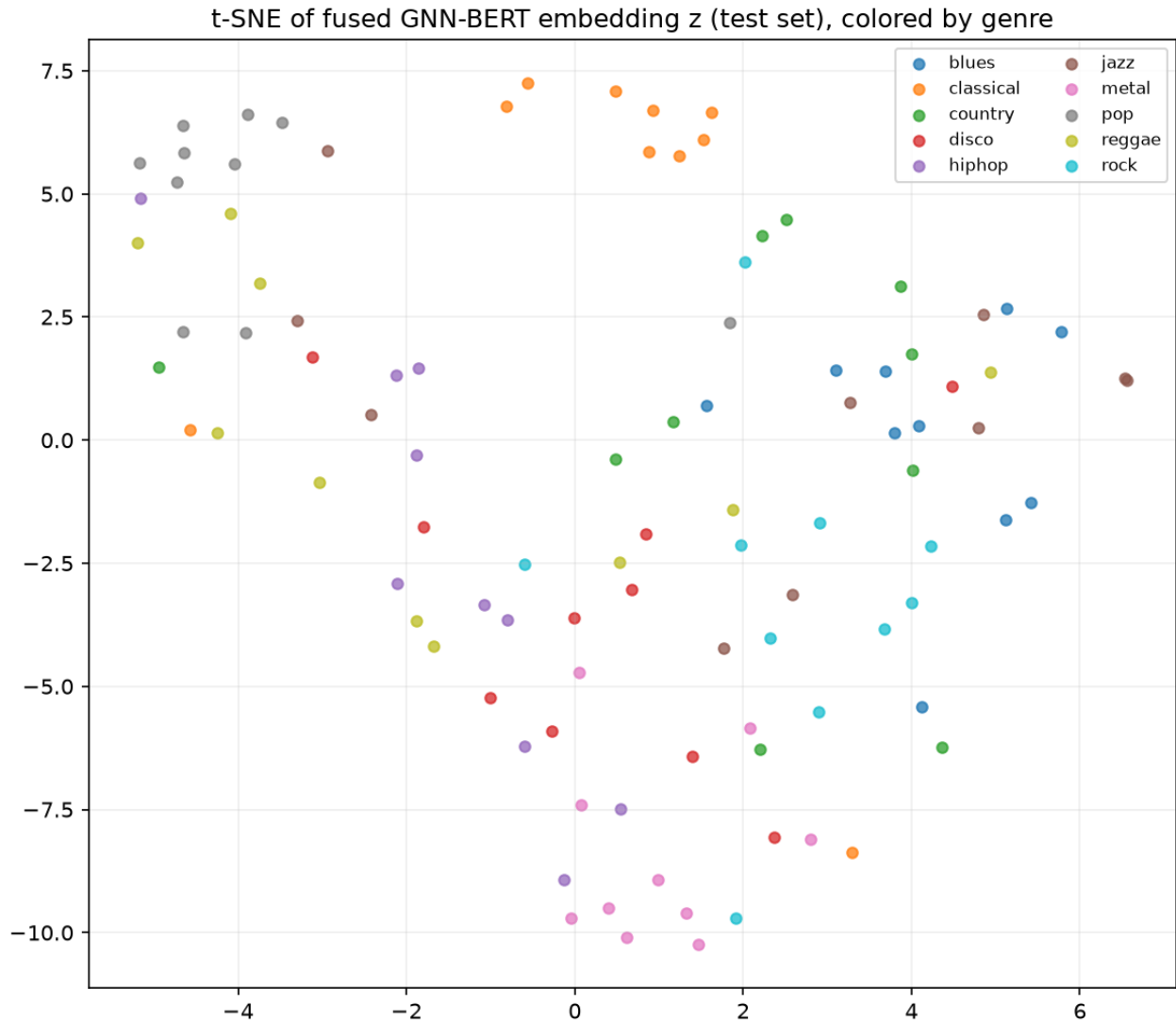


Figure 4: t-SNE of the fused embedding  $z$  on the test set, colored by genre.

## 5.3 Case studies

Case 1 - pop.00002 (true: pop). Top-3 predictions: pop (0.97), reggae (0.02), disco (0.01). Caption: "A fast, energetic recording at about 129 BPM in D minor, with a very bright, treble-heavy timbre and a dense stream of note onsets. Its harmony centres on G minor, F major and D minor, the texture is dominated by percussive attacks, and loudness varies moderately across the clip, with mild noise in the spectrum."

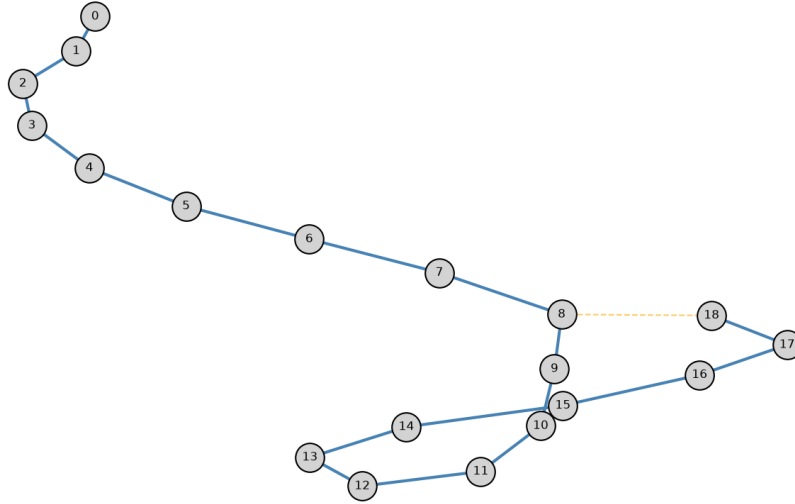
Case 2 - metal.00062 (true: metal). Top-3 predictions: metal (0.94), rock (0.48), disco (0.07). Caption: "A moderately paced recording at about 117 BPM in A minor, with a bright timbre and a dense stream of note onsets. Its harmony centres on A minor, G major and D major, the texture balances sustained harmony with percussive attacks, and loudness stays nearly constant across the clip, with a clean, tonal spectrum."

Case 3 - rock.00010 (true: rock). Top-3 predictions: country (0.67), rock (0.38), blues (0.12). Caption: "A fast, energetic recording at about 161 BPM in D major, with a warm, rounded timbre and a dense stream of note onsets. Its harmony

centres on D major, G major and A major, the texture balances sustained harmony with percussive attacks, and loudness varies moderately across the clip, with a clean, tonal spectrum."

Case study: pop.00002 (true: pop)  
Top-3 predictions: pop (0.97), reggae (0.02), disco (0.01)

Caption: A fast, energetic recording at about 129 BPM in D minor, with a very bright, treble-heavy timbre and a dense stream of note onsets. Its harmony centres on G minor, F major and D minor, the texture is dominated by percussive attacks, and loudness varies moderately across the clip, with mild noise in the spectrum.

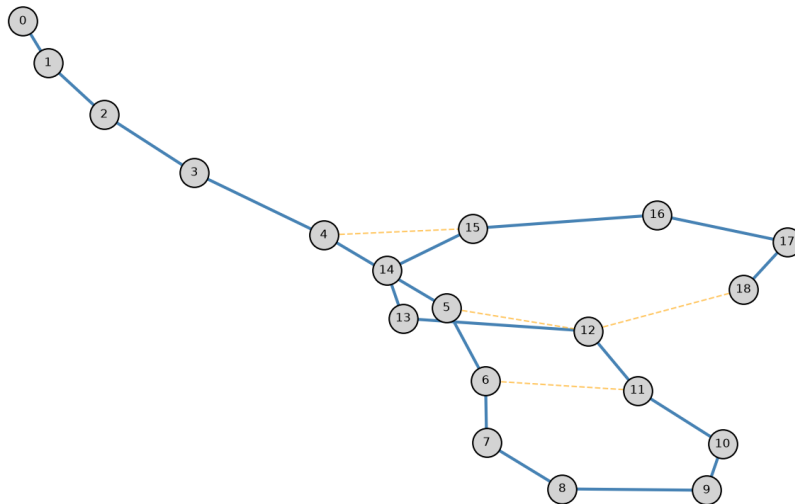


blue solid = temporal adjacency edges | orange dashed = chroma/mel similarity edges

Figure 5: Segment graph for case study 1 (pop.00002). Solid blue = temporal edges; dashed orange = similarity edges.

Case study: metal.00062 (true: metal)  
Top-3 predictions: metal (0.94), rock (0.48), disco (0.07)

Caption: A moderately paced recording at about 117 BPM in A minor, with a bright timbre and a dense stream of note onsets. Its harmony centres on A minor, G major and D major, the texture balances sustained harmony with percussive attacks, and loudness stays nearly constant across the clip, with a clean, tonal spectrum.

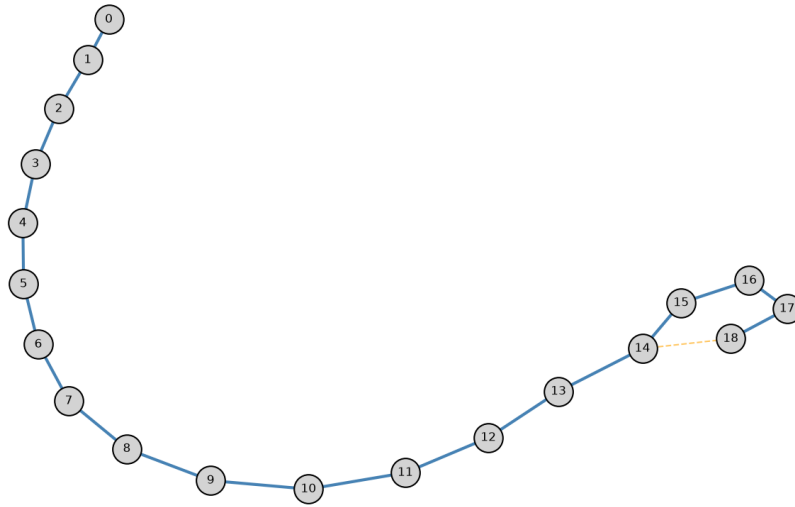


blue solid = temporal adjacency edges | orange dashed = chroma/mel similarity edges

Figure 6: Segment graph for case study 2 (metal.00062). Solid blue = temporal edges; dashed orange = similarity edges.

Case study: rock.00010 (true: rock)  
 Top-3 predictions: country (0.67), rock (0.38), blues (0.12)

Caption: A fast, energetic recording at about 161 BPM in D major, with a warm, rounded timbre and a dense stream of note onsets. Its harmony centres on D major, G major and A major, the texture balances sustained harmony with percussive attacks, and loudness varies moderately across the clip, with a clean, tonal spectrum.



blue solid = temporal adjacency edges | orange dashed = chroma/mel similarity edges

Figure 7: Segment graph for case study 3 (rock.00010). Solid blue = temporal edges; dashed orange = similarity edges.

Qualitative retrieval: across the 10 sampled test captions, 2/10 retrieve their exact paired clip within the top-3, and 2/10 retrieve a same-genre clip at rank 1 - retrieval errors are mostly within-genre confusions, which is the expected failure mode for descriptor-based captions.

## 6 Discussion and Limitations

- Label leakage is the dominant risk in self-captioned datasets: our first pipeline used genre-template captions and reached macro-F1 = 1.0 - a red flag, not a result. Audio-derived captions remove the leak; the resulting scores are honest and clearly lower.
- Graph construction hyperparameters can silently destroy the GNN: non-overlapping 5 s windows with  $\tau = 0.85$  on unstandardized features gave near-complete 6-node graphs (message passing = mean pooling). Overlapping windows, standardized cosine similarity and a top-3 neighbor cap fixed density to  $\sim 0.21$  and visibly improved Task 2.
- GTZAN is single-label, has no artist metadata (so artist-level split hygiene is unenforceable) and has known duplicates/mislabelings; MagnaTagATune (top-50 tags) or FMA would exercise true multi-label behavior with no code changes.
- Chord estimation is naive template matching over chroma, not a trained model; captions are hand-designed descriptors, not human language (MusicCaps is the intended upgrade for Task 4).
- Valence/arousal heads and the NaN-masked emotion loss are implemented but inactive (GTZAN has no emotion labels); adding DEAM would activate them without code changes.

## 7 Conclusion

We delivered the full four-task roadmap on GTZAN: BERT tagging, GraphSAGE structure encoding, cross-attention fusion and contrastive retrieval, evaluated against the two required baselines with validation-tuned thresholds and a single-shot test evaluation. Fusion clearly beats both of its unimodal components and achieves the best probability ranking (AUC-PR) of all models, while a well-tuned spectrogram CNN remains a strong macro-F1 baseline. The project also documents two reproducible failure modes (caption label leakage and degenerate near-complete graphs) whose detection and fixes account for much of the engineering effort - an instructive outcome in itself.

## **References**

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